TEXORA · LECTURE SHEET

Take-up and Let-off Mechanisms

The two motions that feed the loom at its two ends: let-off releases the warp from the warp beam at a controlled, uniform tension, and take-up draws the finished cloth away at exactly the rate the picks are inserted — and it is the take-up, above all, that sets the picks per inch of the cloth.

TE-2103 · WEAVING TECHNOLOGY · TOPIC 9 · LEVEL 2 TERM 1

1. Why these two motions decide cloth quality

Weaving is a balance between two flows of material. At the back, warp must be **let off** — released from the warp beam — at exactly the rate weaving consumes it and always at the same tension, from the full beam to the nearly empty one. At the front, cloth must be **taken up** — drawn forward and wound on — at exactly the rate new cloth forms at the fell. Take-up too slow crowds the picks (cloth heavier and denser than ordered); too fast spreads them (cloth thin and underweight). Uneven let-off swings tension pick to pick, and the cloth shows it at once: tight and loose places, end breaks, wavy edges.

Together with the pick rate, these two motions fix the two quantities a buyer checks first — **picks per unit length** and the **uniformity of warp tension**. This sheet treats take-up first, then let-off, then the faults each produces when it is wrong.

2. The take-up motion

2.1 Purpose and requirements

- To **wind the cloth forward** at the rate picks are inserted, so the cloth fell stays at a fixed position in front of the reed instead of creeping toward the healds.
- To **set the picks per unit length**: for a fixed pick rate, the advance per pick *is* the spacing between successive picks — the take-up motion is the loom's density control.
- To keep that advance **uniform pick after pick** — any variation prints into the cloth as thick and thin places (weft bars).
- To deliver the cloth, under light even tension, to the **cloth roller** for winding on and doffing.

The working element is the **take-up roller** (sand roller / emery roller), surfaced with sand, emery filleting or fine pinning so it grips the cloth without slipping. The cloth passes from the fell over this

roller — a press roller may hold it to the surface — and the roller's slow rotation pulls the cloth sheet forward to the cloth roller, where it is wound on.

2.2 Positive and negative take-up

FEATURE	POSITIVE TAKE-UP	NEGATIVE TAKE-UP
How the take-up roller is driven	Positively, by gears (or a motor) from the loom's motion — every pick advances it by a fixed, calculable amount	Not geared pick-by-pick. The cloth roller is driven by friction, a slipping drive or a weight and simply pulls the cloth; the roller turns because the cloth drags it
Advance per pick	Fixed by the gear train; changed only by changing a wheel (or a setting)	Whatever the pull produces; varies with cloth-roll diameter, friction and tension
Picks per inch	Accurate and repeatable — the standard for quality weaving	Drifts as conditions change; only acceptable for coarse, low-specification work
Typical use	Almost all modern and conventional quality looms	Old and very simple looms; some special coarse fabrics

The rest of this section describes the positive system, in the classic geared form still used to teach — and often still to weave — the principle.

3. The classic multi-wheel take-up train

In the traditional **five-wheel** or **seven-wheel** take-up (the names count the wheels in the train), the motion is taken from the sley, so it is automatically locked to the picking: **one pick, one step of advance**:

1. **The pawl and ratchet.** A rod from the sley works a lever carrying a **pawl** — a small pivoted finger. Each revolution, as the sley swings, the pawl engages a tooth of the **ratchet wheel** and pushes it forward by a fixed number of teeth — commonly one tooth per pick. A retaining (check) pawl stops the wheel slipping back between strokes.
2. **The gear train.** The ratchet shaft carries the first wheel of a spur-wheel train. One wheel is the **change wheel**: swapping it changes the overall reduction, and with it the advance per pick — this is how the weaver sets the picks per inch for a new fabric.
3. **The take-up roller.** The last wheel (often through a worm and worm wheel for the large final reduction) drives the take-up roller, which the heavy reduction turns by only a tiny fraction of a revolution per pick.
4. **Winding on.** The cloth drawn forward is wound onto the cloth roller, driven by friction or a slipping drive so winding tension stays gentle as the roll grows.

Five-wheel and seven-wheel versions differ only in how the reduction is distributed among the wheels; the principle — ratchet stepped once per pick, reduction train, change wheel as the density setting, grippy roller as the output — is identical.

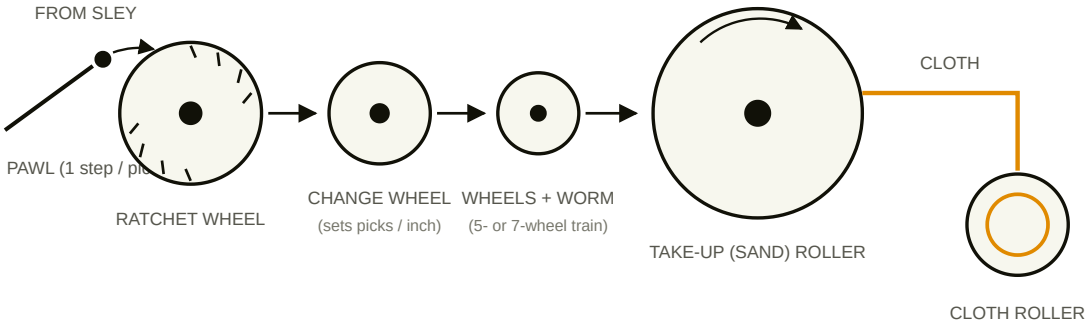


Figure 1 — The positive take-up train, schematically. The sley works the pawl once per pick; the ratchet wheel, change wheel and reduction train convert that step into a minute, exactly repeatable rotation of the take-up roller, which grips and advances the cloth to the cloth roller.

4. The governing relation: from gear train to picks per inch

Because the advance is positive, picks per unit length follow from simple geometry. Per pick the roller turns by a fixed fraction of a revolution, set by the ratchet step and the train's total reduction; the cloth advance per pick is that fraction of the roller's circumference:

$$\text{cloth advance per pick} = \text{take-up roller circumference} \div (\text{effective teeth factor} \times \text{gear reduction})$$

where the denominator is most conveniently used as a single combined number: the number of picks needed to turn the roller once. If the train is arranged so that the roller makes **1 turn in N picks**, then:

$$\text{advance per pick, } a = C \div N \quad \cdot \quad \text{picks per cm} = 1 \div a \quad \cdot \quad \text{picks per inch} = 2.54 \div a \text{ (a in cm)}$$

Two consequences are worth memorising. Picks per inch is **inversely** proportional to advance per pick: double the advance and the picks per inch halve. And the change wheel acts through N: a wheel that turns the roller faster per pick (smaller N) gives a coarser cloth; one that slows it (larger N) a denser cloth — the shop-floor rule "more picks wanted, slower take-up".

> **WORKED EXAMPLE — PICKS PER INCH FROM THE TAKE-UP TRAIN**
>
> A take-up roller has a circumference of 45 cm. The ratchet, change wheel and reduction train together make the roller advance by 1/600 of a turn per pick. Find the picks per inch of the cloth.
>
> Advance per pick, a = 45 cm ÷ 600 = **0.075 cm per pick**.

Picks per cm = 1 ÷ 0.075 = **13.33 picks/cm**.

Picks per inch = 13.33 × 2.54 = **33.9 picks/inch** (33.86 before rounding).

Check by the direct route: picks per inch = 2.54 ÷ 0.075 = 33.87 ✓ — the two routes agree.

WORKED EXAMPLE — EFFECT OF CHANGING THE CHANGE WHEEL

The change wheel is swapped for one that makes the same 45 cm roller advance by 1/540 of a turn per pick. Find the new picks per inch.

Advance per pick, a = 45 cm ÷ 540 = **0.0833 cm per pick**.

Picks per cm = 1 ÷ 0.0833 = **12.0 picks/cm**.

Picks per inch = 12.0 × 2.54 = **30.5 picks/inch**.

As the relation predicts: a larger advance per pick spreads the same picks over more cloth, and density falls from 33.9 to 30.5 picks/inch.

4.1 Continuous and electronic take-up

Modern looms increasingly use a **continuous take-up**: the roller is driven smoothly by its own motor, electronically synchronised to the pick rate. In a fully **electronic take-up** the required picks per unit length is typed in at the panel; there is no change wheel, and density can even be varied within a fabric (a denser border, a programmed pick pattern). The governing relation above is unchanged — only the mechanism enforcing it is electronic.

5. The let-off motion

5.1 Purpose and requirements

- To **release warp from the warp beam** at exactly the rate weaving consumes it — no faster (slack warp), no slower (over-tight warp).
- To hold **warp tension uniform** — over time (full beam to empty), across the width, and within each cycle, absorbing the peaks of shed opening and beat-up.
- To do this automatically as the beam empties, although a shrinking beam delivers less warp per revolution and presents a smaller lever arm to any brake.

From the beam the warp passes over the **back rest** — a roller (fixed, or spring-loaded and free to yield) at the back of the loom — which turns the warp into the horizontal warp line toward the healds and absorbs the cycle's sharp tension peaks. The back rest is guide, tension cushion, and — in electronic systems — the tension *sensor*.

5.2 Negative let-off: friction control

In a **negative let-off** the beam is **not driven at all**. Warp tension itself pulls the beam round against a friction brake: a rope or chain passed around the **ruffle** (the grooved flange at the end of the beam barrel), one end anchored, the other loaded with weights, directly or through a lever. The beam turns only when tension overcomes the friction — so warp is delivered at the tension the brake setting imposes.

- **Setting:** more weight (or a longer lever ratio) → more friction → higher warp tension.
- **The diameter problem:** braking is a fixed retarding *moment*, while tension acts at the beam's surface radius: tension = brake moment ÷ radius. As the beam empties and its radius falls, **tension rises** unless the weaver progressively removes weight — this hand compensation is the friction let-off's classic weakness.
- **Character:** simple, cheap, robust; but tension depends on the weaver's attention, rope condition and beam diameter — suited to coarse and medium work.

5.3 Positive let-off: geared drive

In a **positive let-off** the warp beam is **driven** — geared from the loom (worm drive, change gearing or a variable-speed unit) so it positively delivers warp. Because length delivered per revolution falls as the beam empties, a fixed ratio is not enough: the drive is governed by tension itself. Classically the yielding back rest is linked to the drive (or to a friction clutch / variable pulley) so a rise in tension speeds the beam slightly and a fall slows it — a mechanical feedback loop holding tension constant at any diameter. Positive let-off is the standard on better looms.

5.4 Electronic let-off

A sensor measures back-rest position (that is, warp tension) continuously and a servo motor drives the beam, adjusting many times per second to hold the set tension and compensating automatically for beam diameter. Tension is set digitally and reproduced exactly for the next beam — the let-off equivalent of the electronic take-up of §4.1; the two are usually fitted together.

FEATURE	NEGATIVE (FRICTION) LET-OFF	POSITIVE (GEARED / ELECTRONIC) LET-OFF
Beam driven?	No — pulled round by warp tension against a brake	Yes — driven by gearing or a servo motor
Tension set by	Weights / lever on the rope or chain over the ruffle	Drive control, governed by back-rest feedback; digitally in electronic form
As beam empties	Tension rises unless weights are reduced by hand	Compensated automatically

Tension uniformity	Moderate; depends on weaver and rope condition	High and repeatable

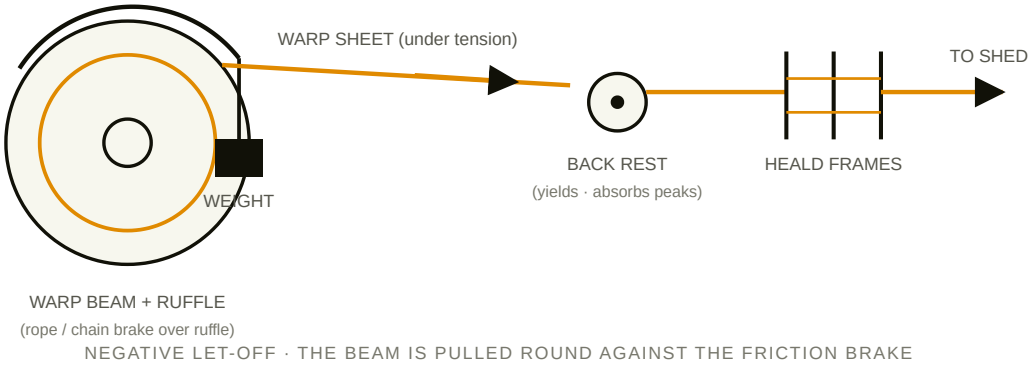


Figure 2 — Negative (friction) let-off, side view. The warp sheet pulls the beam round against the rope-and-weight brake acting on the ruffle; the sheet then passes over the yielding back rest to the heald frames. As the beam empties, the same brake moment acts at a smaller radius, so tension rises unless weight is removed.

6. Faults from the two motions

FAULT IN CLOTH OR WEAVING	MOTION RESPONSIBLE	USUAL CAUSE
Variable picks per inch along the piece (density bands, weft bars)	Take-up	Worn or damaged change wheel / ratchet teeth; pawl missing steps; slack or worn train; cloth slipping on a smooth or damaged take-up roller surface
Whole piece too dense or too open against the order	Take-up	Wrong change wheel fitted, or wrong density set on an electronic take-up
Tight warp — frequent end breaks, harsh weaving	Let-off	Excess brake weight; beam not compensated as it empties; back rest seized so peaks are not absorbed
Loose / slack warp — slack ends, loops, bad shed	Let-off	Too little brake weight; rope slipping on the ruffle; positive drive set fast
Wavy cloth, wavy edges, tension streaks	Let-off (and beam state)	Uneven brake action side to side; unevenly wound beam; damaged ruffle; back rest bent or unevenly loaded
Start-up marks (dense / open band after a loom stop)	Both	Fell moves while the loom stands (take-up creep, warp stretching / relaxing); modern looms inch take-up / let-off at restart to correct it

Diagnostic rule: a fault repeating **with the pick spacing** points to take-up; a fault following **tension** — breaks, slackness, waviness — points to let-off.

Glossary

Take-up motion — draws woven cloth forward and winds it on, at the rate picks are inserted. **Let-off motion** — releases warp from the warp beam at controlled tension. **Positive take-up / let-off** — the element is driven by gearing or motor, so delivery is fixed and calculable. **Negative take-up / let-off** — delivery results from pull against friction or a slipping drive. **Ratchet wheel and pawl** — toothed wheel and engaging finger converting the sley's swing into one fixed step of rotation per pick. **Change wheel** — the interchangeable train wheel whose size sets picks per unit length. **Take-up (sand) roller** — the rough-surfaced roller that grips and advances the cloth. **Ruffle** — the grooved beam flange around which the brake rope or chain passes. **Back rest** — the rear roller guiding the warp into the warp line and absorbing tension peaks. **Cloth fell** — the line where the last pick was beaten up and cloth is forming.

Key points

• Take-up winds the cloth on at the rate picks are inserted — and thereby sets the picks per unit length; let-off releases warp at controlled, uniform tension.
• In the classic positive take-up, a pawl steps a ratchet wheel once per pick and a 5- or 7-wheel train (with its change wheel) reduces that step to a minute fraction of a turn of the sand roller.
• Advance per pick = roller circumference ÷ picks per roller turn; picks per inch = 2.54 ÷ advance in cm — density is inversely proportional to advance, so more picks wanted means slower take-up.
• Negative let-off brakes the beam with a weighted rope over the ruffle; tension rises as the beam empties unless weight is removed. Positive and electronic let-off drive the beam under back-rest feedback and compensate automatically.
• The back rest guides the warp, absorbs the tension peaks of shedding and beat-up, and in electronic systems senses the tension.
• Pick-spacing faults come from take-up; tension faults — breaks, slack ends, wavy cloth — come from let-off.

Practice

1. A take-up roller of circumference 40 cm advances by 1/500 of a turn per pick. Calculate the cloth advance per pick, the picks per cm and the picks per inch.
2. A fabric is ordered at 20 picks/cm on a loom whose take-up roller circumference is 48 cm. What advance per pick is required, and what fraction of a roller turn per pick must the train give?

3. A warp beam delivers warp by its surface. Calculate the warp length released per beam revolution when the beam diameter is 50 cm and when it has emptied to 25 cm, and use your results to explain why a fixed-ratio positive drive cannot hold tension without feedback.
4. Explain why, in a friction let-off, the warp tension increases as the beam empties if the brake weights are left unchanged, and state what the weaver must do during the beam's life.
5. A piece of cloth shows bands of higher and lower pick density repeating irregularly along its length, while warp tension and end-break rate are normal. Which motion do you inspect first, and which three components do you check?
6. State two advantages of electronic take-up and two of electronic let-off over their geared and friction predecessors.

Answer key

1. Advance = 40 ÷ 500 = **0.08 cm per pick**. Picks per cm = 1 ÷ 0.08 = **12.5**. Picks per inch = 12.5 × 2.54 = **31.75 ≈ 31.8 picks/inch**.
2. Advance per pick = 1 ÷ 20 = **0.05 cm**. Fraction of a turn = 0.05 ÷ 48 = 1/960, so the train must give **1/960 of a turn per pick** (one roller turn per 960 picks).
3. Length per revolution = π × D. At D = 50 cm: 3.1416 × 50 = **157.1 cm/rev**. At D = 25 cm: 3.1416 × 25 = **78.5 cm/rev** — exactly half. A fixed ratio turns the beam at a fixed rate, so delivery per minute would halve as the beam empties while consumption stays constant, and tension would climb continuously. Back-rest (or sensor) feedback must speed the beam up as its diameter falls.
4. The brake applies an almost constant retarding moment to the beam, while the warp pulls at the beam's surface. Tension = brake moment ÷ beam radius. As the beam empties, the radius falls, so the same moment balances a larger tension — the warp gets progressively tighter. The weaver must progressively remove weight (or ease the lever) as the beam diameter falls, checking tension as the beam runs down.
5. Inspect the **take-up** first — the fault is in pick spacing while tension is normal. Check: (i) the pawl and ratchet for missed or double steps and worn teeth; (ii) the change wheel and train for damaged teeth or slack; (iii) the take-up roller surface and cloth path for slipping (worn sand/emery covering, oily cloth, insufficient wrap or press).
6. Electronic take-up: density set digitally, no change wheel (fast, exact changeovers), and variable within a fabric by programme. Electronic let-off: tension held constant automatically, full beam to empty, by sensor feedback, and stored for exact reproduction on repeat orders.
